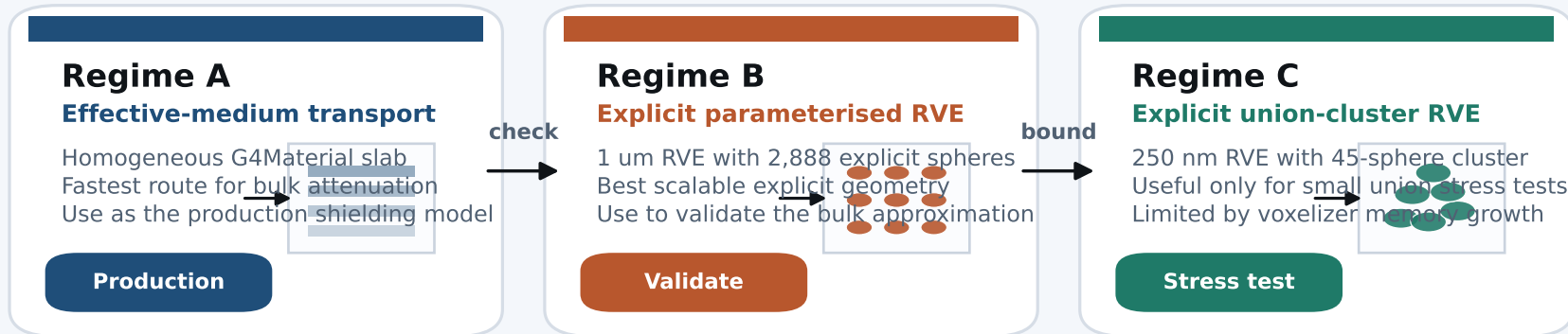


Graphical Abstract: Geant4 Modelling Regimes for Nanocomposite Shielding

Matched-composition HDPE/Bi₂O₃ benchmark: production model, explicit validation RVE, and union stress test.



Main result and workflow decision

All three regimes stayed within the pre-registered 3% agreement window at 30-150 keV.

Recommended workflow: A for production attenuation, B for explicit validation, and C only for small union stress tests.

0.549% mean error

< 3% deviation

2,888 spheres feasible

~45 spheres on 8 GB